Constitutional Semantic Hashing: Governed Commitment Schemes for Semantic Provenance, Admissibility, and Constitutional Continuity

Commitment Relations Over Declared Interpretation Regimes Rather Than Structural Identity Alone

Constitutional Semantic Hashing: Governed Commitment Schemes for Semantic Provenance, Admissibility, and Constitutional Continuity asks what meaning must preserve before it can govern what follows.

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Abstract

Cryptographic systems traditionally secure integrity, authenticity, confidentiality, and provenance at the level of bytes, messages, or formally structured commitments. However, governance-bound semantic systems increasingly require stronger guarantees: determining whether artifacts remain equivalent under declared interpretation regimes, whether continuity claims remain admissible under evolving governance constraints, and whether lineage, authority, and semantic interpretation remain constitutionally continuous across system evolution.

This paper introduces **Constitutional Semantic Hashing (CSH)**, a governed commitment framework that composes existing audited cryptographic primitives—including BLAKE3, SHA-256, Ed25519, deterministic canonicalization schemes, and lineage-aware receipt chains—with explicitly declared semantic interpretation regimes.

CSH is not a novel cryptographic primitive. It is a governed composition framework providing semantics- and governance-aware commitment relations over canonicalized representations using existing cryptographic tools.

The central claim is:

> Hashes become substantially more operationally useful when interpreted as commitments over declared equivalence classes under specified governance regimes rather than merely byte-level structure.

Under CSH:

- interpretation regimes become explicit governance objects,

- admissibility becomes part of identity,
- authority becomes cryptographically bound to interpretation context,
- lineage becomes part of the commitment boundary,
- and continuity becomes conditional upon declared governance constraints.

This enables:

- governed schema evolution,
- deterministic divergence detection relative to declared regimes,
- provenance-aware identity,
- governance-bound receipt chains,
- and admissibility-aware continuity management.

CSH is deployable as a thin governance layer over existing cryptographic stacks, enabling systems such as MCIR, CONDEX, CMB, and DKN to enforce deterministically verifiable continuity and divergence detection without modifying their underlying cryptographic primitives.

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1. Introduction

Conventional cryptographic systems answer structural questions:

- *Are these bytes identical?*
- *Was this message modified?*
- *Did this signer authorize this payload?*
- *Does this commitment verify?*

These are foundational capabilities.

However, governance-bound semantic systems increasingly require stronger guarantees:

- *Did anything governance-relevant change under the declared interpretation regime?*
- *Does this schema preserve admissible continuity?*
- *Was this artifact produced under the same authority and law context?*
- *Does this lineage preserve constitutional continuity?*
- *Is this still the same governed commitment under the declared regime?*

Traditional hashes cannot answer these questions because:

> structural identity and governance-relative identity are not equivalent.

Two artifacts may differ structurally while remaining continuity-equivalent under a declared interpretation regime:

- field ordering changes,
- serialization changes,
- rendering migrations,
- schema refactors,

- or naming normalization.

Conversely, two structurally identical artifacts may diverge constitutionally:

- authority changes,
- admissibility-law changes,
- interpretation-regime changes,
- lineage discontinuities,
- or governance-scope mutations.

Constitutional Semantic Hashing addresses this gap by treating:

> interpretation regimes themselves as first-class governance objects.

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2. Relationship to Semantic Hashing

Semantic Hashing established the core maneuver:

1. Select the distinctions willing to survive.
2. Canonicalize the surviving representation.
3. Hash the canonical form.
4. Treat the resulting hash as a commitment to an equivalence class under the declared lens.

This reframed hashing from:

> structural integrity checking

into:

> governed equivalence-class commitment.

CSH extends this structure by incorporating:

- authority context,
- admissibility-law context,
- lineage continuity,
- artifact provenance class,
- and interpretation-regime identity

into the commitment relation itself.

The result is not:

> stronger cryptography,

but:

> richer governance-aware commitment semantics over existing cryptographic primitives.

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3. Definitions

3.1 Artifact

An artifact is a structured representation participating in a governance-aware semantic system.

Examples include:

- receipts,
- policies,
- schemas,
- semantic claims,
- MCIR objects,
- CONDEX artifacts,
- or governance transitions.

Artifacts are always interpreted relative to:

- an artifact class,
- a declared interpretation regime,
- and an admissibility context.

3.2 Interpretation Regime

An interpretation regime defines:

- what distinctions are relevant,
- what transformations preserve continuity,
- what normalization rules apply,
- and what equivalence boundaries govern admissibility.

An interpretation regime includes:

- a semantic lens,
- a canonicalization procedure,
- a governance context,
- and a versioned admissibility structure.

Identity under CSH is always:

> regime-relative rather than universal.

3.3 Semantic Lens

A semantic lens is a deterministic projection from an artifact into a canonical representation preserving only governance-relevant distinctions.

The lens defines:

- what survives,
- what is discarded,
- and what constitutes equivalence under the regime.

The lens is therefore:

> an operational interpretation boundary rather than a philosophical semantic claim.

3.4 Admissibility Law

An admissibility law defines:

- which commitments are permitted,
- which transitions preserve continuity,
- and which equivalence relations are constitutionally valid.

Admissibility laws are:

- explicit,
- versioned,
- hashable,
- governance-bound,
- and distributable through signed publication mechanisms such as registries or constitutional ledgers.

The admissibility law therefore becomes:

> a discoverable governance artifact participating directly in continuity verification.

3.5 Authority

Authority identifies the entity authorized to produce or validate a governed commitment within a specified admissibility regime.

Authority is interpreted relative to:

- the regime,
- the artifact class,
- and the lineage context.

3.6 Lineage

Lineage is the ordered continuity relation connecting governed commitments through admissible rebinding transitions.

Lineage is part of the commitment relation itself rather than external metadata.

Both:

- forward continuity, and:
- backward continuity

are represented through chained `prior_receipt_hash` relations.

Lineage replay therefore becomes:

> deterministic traversal of admissible continuity links under the declared regime.

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4. Constitutional Semantic Hashes

A Constitutional Semantic Hash is defined as:

```
CSH = H(
  algorithm_id,
  interpretation_regime_id,
  regime_version,
  authority_pubkey,
  admissibility_law_hash,
  artifact_type,
  canonical_semantic_form,
  prior_receipt_hash
)
```

Where:

Field	Purpose
`algorithm_id`	Audited primitive used
`interpretation_regime_id`	Declared equivalence regime
`regime_version`	Interpretation regime version
`authority_pubkey`	Authority exercising the commitment
`admissibility_law_hash`	Governance regime governing admissibility
`artifact_type`	Semantic provenance class
`canonical_semantic_form`	Normalized governance-relevant representation
`prior_receipt_hash`	Lineage continuity anchor

The underlying primitive may be:

- BLAKE3,
- SHA-256,
- SHA-3,
- or another audited cryptographic primitive.

CSH does not modify these primitives.

It governs:

- commitment construction,
- interpretation context,
- admissibility boundaries,
- and continuity semantics.

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5. Commitment Relations

CSH is best understood as:

> a governed commitment scheme over a declared canonicalization and governance regime.

Define a commitment:

```
c = (
  regime,
  authority,
  law,
  lineage,
  canonical_form,
  h
)
```

such that:

```
h = CSH(...)
```

Two commitments are continuity-equivalent under a declared regime iff:

$$h_1 = h_2$$

within:

- the same interpretation regime,
- admissibility law,
- artifact class,
- and lineage assumptions.

Hash equality therefore means:

> continuity under declared governance conditions.

Not:

> universal semantic sameness.

This distinction is foundational.

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6. Canonicalization as Governance

CSH depends fundamentally on deterministic canonicalization.

Without deterministic canonicalization:

- equivalence classes fragment,
- replayability fails,
- and continuity becomes unstable.

JSON Canonicalization Scheme (JCS, RFC 8785) provides a strong baseline for structured serialization. However, governance-bound semantic systems frequently require:

- governance-relevant-field selection,
- semantic normalization,
- admissibility-aware collapse,
- and interpretation-specific equivalence handling.

Canonicalization therefore becomes:

> executable governance interpretation.

The interpretation regime determines:

- what distinctions survive,
- what distinctions are denied relevance,
- and what transformations preserve continuity.

This is not merely formatting.

It is:

> governed semantic reduction under declared admissibility constraints.

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7. Constitutional Continuity and Divergence

Traditional hashes identify:

> structural sameness.

CSH identifies:

> continuity under a declared interpretation and governance regime.

This means:

- formatting changes,
- field reordering,
- serialization migration,

- or naming normalization

may preserve constitutional continuity under one regime.

Conversely:

- authority transfer,
- admissibility-law mutation,
- interpretation-regime changes,
- or lineage discontinuity

may produce constitutional divergence even when structure remains unchanged.

CSH therefore does not claim:

> objective semantic identity.

It claims:

> governance-relative continuity under declared interpretation constraints.

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8. Deterministic Divergence Detection

Most semantic systems detect divergence heuristically:

- embeddings,
- similarity scoring,
- narrative interpretation,
- or human review.

CSH enables deterministic divergence detection relative to a declared interpretation regime.

If:

$$\mathrm{CSH}_{\mathrm{new}} = \mathrm{CSH}_{\mathrm{prior}}$$

then:

> no governance-relevant distinction changed under the declared regime.

If:

$$\mathrm{CSH}_{\mathrm{new}} \neq \mathrm{CSH}_{\mathrm{prior}}$$

then:

> a governance-relevant distinction diverged under the declared regime.

This distinction is intentionally scoped.

CSH does not claim:

> universal semantic drift detection.

It claims:

> deterministic divergence detection relative to explicit equivalence boundaries.

This transforms:

- silent reinterpretation,
- schema laundering,
- hidden authority substitution,
- and continuity mutation

into mechanically observable state transitions.

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9. Threat Model and Governance Assumptions

CSH assumes:

- authenticated authority keys,
- deterministic canonicalization,
- explicit interpretation-regime declaration,
- signed regime definitions,
- and governed admissibility-law management.

The framework further assumes minimal trust in:

- regime operators,
- governance authorities,
- and canonicalization publishers.

Primary threats include:

- spoofed interpretation regimes,
- malicious canonicalization ambiguity,
- governance capture,
- authority substitution,
- weak equivalence boundaries,
- semantic over-collapse,
- lineage forgery,
- and regime-discovery attacks.

Mitigations include:

- signed regime publication,
- versioned regime discovery,

- multi-party governance approval,
- deterministic replay verification,
- and explicit admissibility-law lineage.

CSH provides:

> a mechanism for enforcing governance-relative continuity commitments.

It does not guarantee:

- correctness of governance regimes,
- political legitimacy,
- incentive alignment,
- or semantic appropriateness of the declared interpretation systems.

These are governance failures rather than failures of the underlying cryptographic primitives.

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10. Worked Example

Consider two governance artifacts:

```
{
  "policy_scope": "district",
  "max_energy_draw": 1200
}
```

and:

```
{
  "scope": "district",
  "max_energy_draw": 1200
}
```

Under:

- a structural hash, the identities diverge.

Under:

- an interpretation regime treating `policy_scope` and `scope` as equivalent governance labels, the canonical semantic forms converge.

The resulting CSH values remain identical if:

- authority remains unchanged,
- admissibility law remains unchanged,
- lineage continuity remains admissible,
- and interpretation regime remains unchanged.

However, if:

- the authority changes,
- the admissibility law changes,
- or the interpretation regime changes,

then constitutional continuity diverges even when canonical structure remains identical.

Identity therefore depends on:

> governed commitment context, not merely structural representation.

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11. Comparison to Related Approaches

- **Approach:** Traditional Hashing
  Structural Integrity: Yes
  Governance Regime: No
  Authority Context: No
  Lineage Continuity: No
  Cannot Express: Governance-relative continuity

- **Approach:** Signed Metadata
  Structural Integrity: Partial
  Governance Regime: Partial
  Authority Context: Yes
  Lineage Continuity: No
  Cannot Express: Interpretation-bound equivalence

- **Approach:** Content Addressing
  Structural Integrity: Yes
  Governance Regime: No
  Authority Context: No
  Lineage Continuity: No
  Cannot Express: Admissibility-aware identity

- **Approach:** Semantic Hashing
  Structural Integrity: Relative to Lens
  Governance Regime: Partial
  Authority Context: No
  Lineage Continuity: No
  Cannot Express: Governance-bound lineage

- **Approach:** CSH
  Structural Integrity: Relative to Declared Regime
  Governance Regime: Yes
  Authority Context: Yes
  Lineage Continuity: Yes
  Cannot Express: N/A

CSH differs from:

- content-addressing,
- signed metadata,
- blockchain commitments,
- and ordinary semantic hashing

because the commitment relation itself includes:

- interpretation regime,
- admissibility law,
- authority context,
- and lineage continuity.

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12. Protocol Sketch

A minimal CSH workflow proceeds as follows:

1. Publish signed interpretation regime definition.
2. Publish signed admissibility-law definition.
3. Canonicalize artifact under the regime.
4. Construct canonical semantic form.
5. Bind:
    - regime,
    - authority,
    - admissibility law,
    - artifact class,
    - lineage anchor, into the commitment tuple.
6. Compute CSH using audited primitive.
7. Sign and publish receipt.
8. Verify continuity through deterministic replay.

This allows:

- replayable governance verification,
- deterministic divergence detection,
- admissibility-aware lineage traversal,
- and governance-relative continuity validation.

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13. MCIR and Governance-Aware Identity

CSH composes naturally with MCIR architectures.

Under MCIR:

- artifacts possess typed constitutional meaning,
- transitions are governed,

- admissibility is explicit,
- and continuity is regulated constitutionally.

The MCIR identifier of an artifact therefore becomes:

> its Constitutional Semantic Hash under the declared interpretation regime.

Identity tracks:

- admissible participation continuity, rather than:
- representation structure alone.

This allows:

- governed schema evolution,
- deterministic continuity verification,
- semantic receipt chains,
- governance-aware replayability,
- and constitutional memory continuity.

CSH may additionally compose with:

- propositional admissibility systems,
- temporal governance logic,
- and higher-order policy verification frameworks

to express richer continuity and admissibility properties over semantic systems.

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14. Conclusion

Traditional cryptographic systems secure:

- structural integrity,
- signatures,
- and message authenticity.

Semantic Hashing secures:

- declared equivalence classes under explicit lenses.

Constitutional Semantic Hashing extends this into governance-aware systems by incorporating:

- interpretation regimes,
- admissibility laws,
- authority context,
- provenance class,
- and lineage continuity

into the commitment relation itself.

The result is not:

> stronger cryptographic primitives,

but:

> governance-aware commitment schemes over existing audited primitives.

CSH enables systems to distinguish:

- structural change, from:
- governance-relevant divergence under declared interpretation regimes.

As governance-aware semantic systems, constitutional memory architectures, provenance-bound AI systems, and admissibility-governed infrastructures become increasingly important, frameworks like Constitutional Semantic Hashing provide a mechanically verifiable foundation for regime-relative continuity, deterministic divergence detection, provenance-aware identity, and admissibility-aware governance.

Corpus-Wide Reference Positioning

This paper argues that cryptographic systems traditionally secure integrity, authenticity, confidentiality, and provenance at the level of bytes, messages, or formally structured commitments. Its nearest external comparison fields are crypto, semantic, governance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A valid hash, signature, or inclusion proof establishes a bounded cryptographic relation, not meaning, authority, safety, or jurisdiction by itself. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. These limits are not

generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Anders Rundgren, Bret Jordan, and Samuel Erdtman. *JSON Canonicalization Scheme (JCS)*. RFC 8785, 2020. https://www.rfc-editor.org/rfc/rfc8785 **Relationship:** Adjacent canonicalization standard relevant to stable cryptographic commitments.

[**E2**] Ben Laurie et al. *Certificate Transparency Version 2.0*. RFC 9162, 2021. https://www.rfc-editor.org/rfc/rfc9162 **Relationship:** Adjacent append-only Merkle-log standard for inclusion and consistency evidence.

[**E3**] Manu Sporny et al., eds. *Decentralized Identifiers (DIDs) v1.0*. W3C Recommendation, 19 July 2022. https://www.w3.org/TR/did-core/ **Relationship:** Adjacent decentralized identity standard; identifier resolution is not assumed to settle semantic standing or jurisdiction.

[**E4**] Juan Benet. "IPFS - Content Addressed, Versioned, P2P File System." 2014. arXiv:1407.3561. https://arxiv.org/abs/1407.3561 **Relationship:** Predecessor content-addressed network architecture; content identity is compared with, not treated as sufficient for, semantic jurisdiction.

[**E5**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E6**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E7**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012.

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Ableman corpus

[**A1**] Adam Ableman. *The Three Hash Rule.* Ableman Corpus current edition; work identity `PAPER.PUB.20788617`;
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[**A3**] Adam Ableman. *Self-Certifying Proof-Carrying Identity for Registrar-Free Namespace Networks.* Ableman Corpus current edition; work identity `PAPER.PUB.20114888`;
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[**A4**] Adam Ableman. *Semantic Hashing: Hashes as Declared Equivalence Classes of Meaning.* Ableman Corpus current edition; work identity `PAPER.PUB.18500533`;
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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DOI: 10.5281/zenodo.20248734